CSE 270 Sports Analytics

Predicting with distributions

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Predicting with distributions

Predicting with distributions

Suppose we have two variables with normal distribution,

- Home team score
- Away team score

$$H \sim N(\mu_H, \sigma_H^2)$$

$$A \sim N(\mu_A, \sigma_A^2)$$

Then the difference between the home and away teams scores will also have normal distribution with the following parameters (assuming this two are independent)

$$\mu_d = \mu_H - m_A$$
$$\sigma_D^2 = \sigma_H^2 + \sigma_A^2$$

Thus the difference will have a normal distribution with the following parameters

$$D \sim N(\mu_H - \mu_A, \sigma_H^2 + \sigma_A^2)$$

Predicting with distributions

- In NBA (National Basketball Association) points scored per game by home and away teams follow normal distribution.
- Install package SportsAnalytics270 from github.
- The package contains data for Sports analytics.

```
install.packages("remotes")
remotes::install_github("HABET/CSE270", dependencies = TRUE)
SportsAnalytics270::install_github_course_packages(force=T)
```

Predicting with distributions

The list of necessary packages

```
library(SportsAnalytics270)
library(dplyr)
library(ggplot2)
library(reshape2)
library(zoo)
library(VGAM)
```

Predicting with distributions

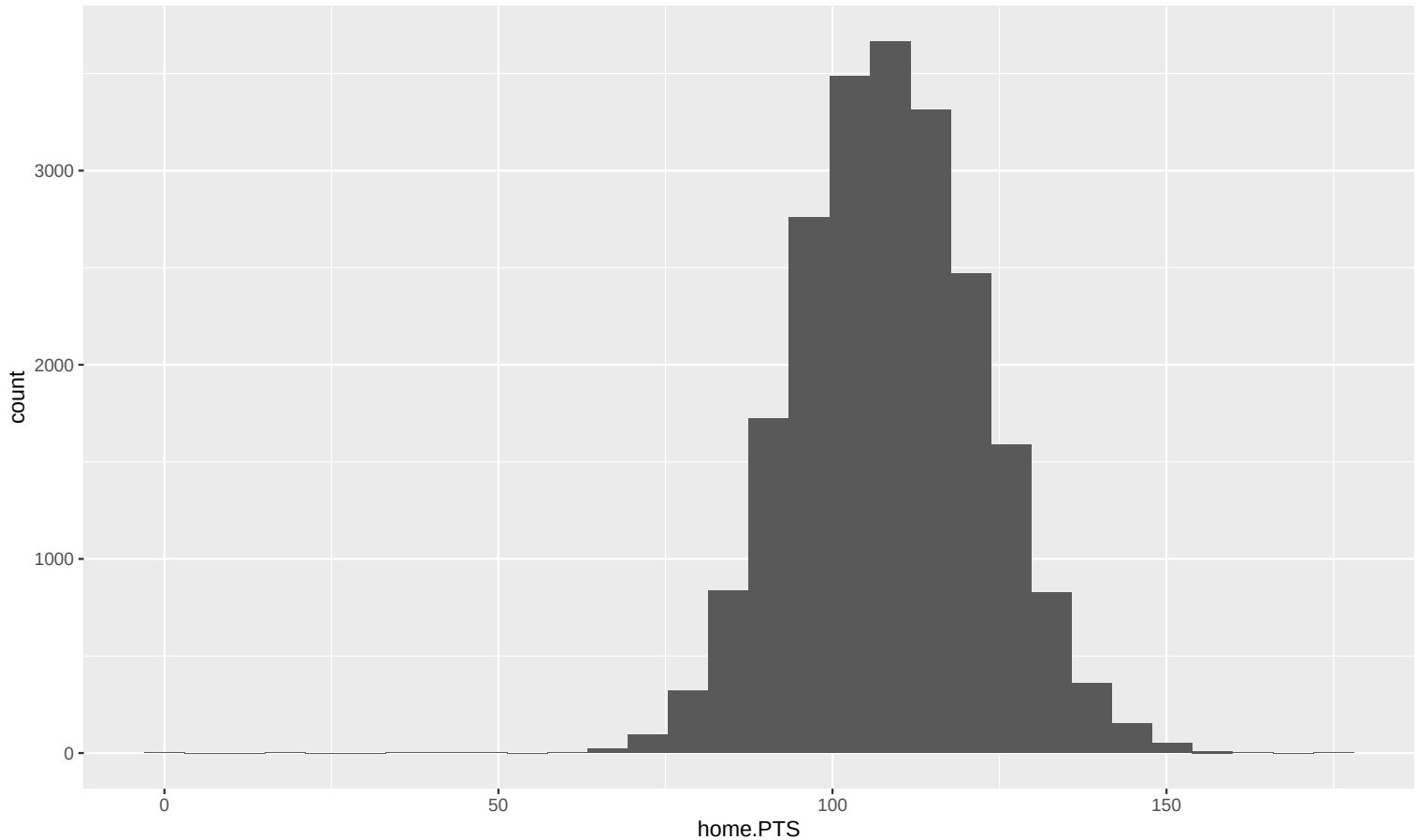
Dataframe nba_games contains NBA regular season games from 2009 to 2025-2026 season

```
options(width = 65)
data("nba_games")
head(nba_games, 2)
##   SEASON_ID  GAME_DATE home.TEAM_ABBREVIATION
## 1      2009 2009-10-27                    CLE
## 2      2009 2009-10-27                    DAL
##        home.TEAM_NAME home.PTS away.TEAM_ABBREVIATION
## 1 Cleveland Cavaliers       89                    BOS
## 2    Dallas Mavericks       91                    WAS
##       away.TEAM_NAME away.PTS home.WL
## 1     Boston Celtics       95       L
## 2 Washington Wizards      102       L
```

Predicting with distributions

Plot the histogram for home points

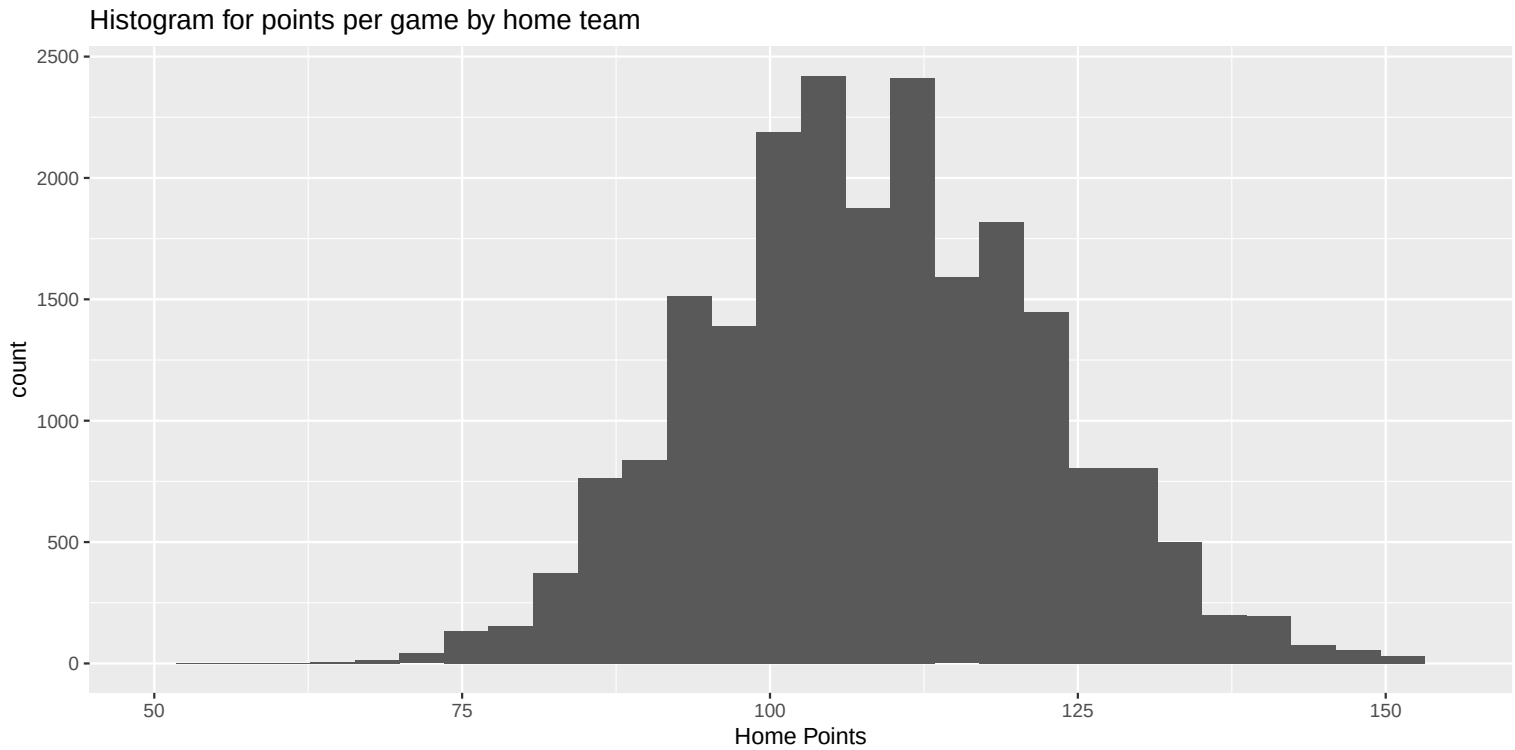
```
ggplot(nba_games, aes(x=home.PTS))+geom_histogram()
```



Predicting with distributions

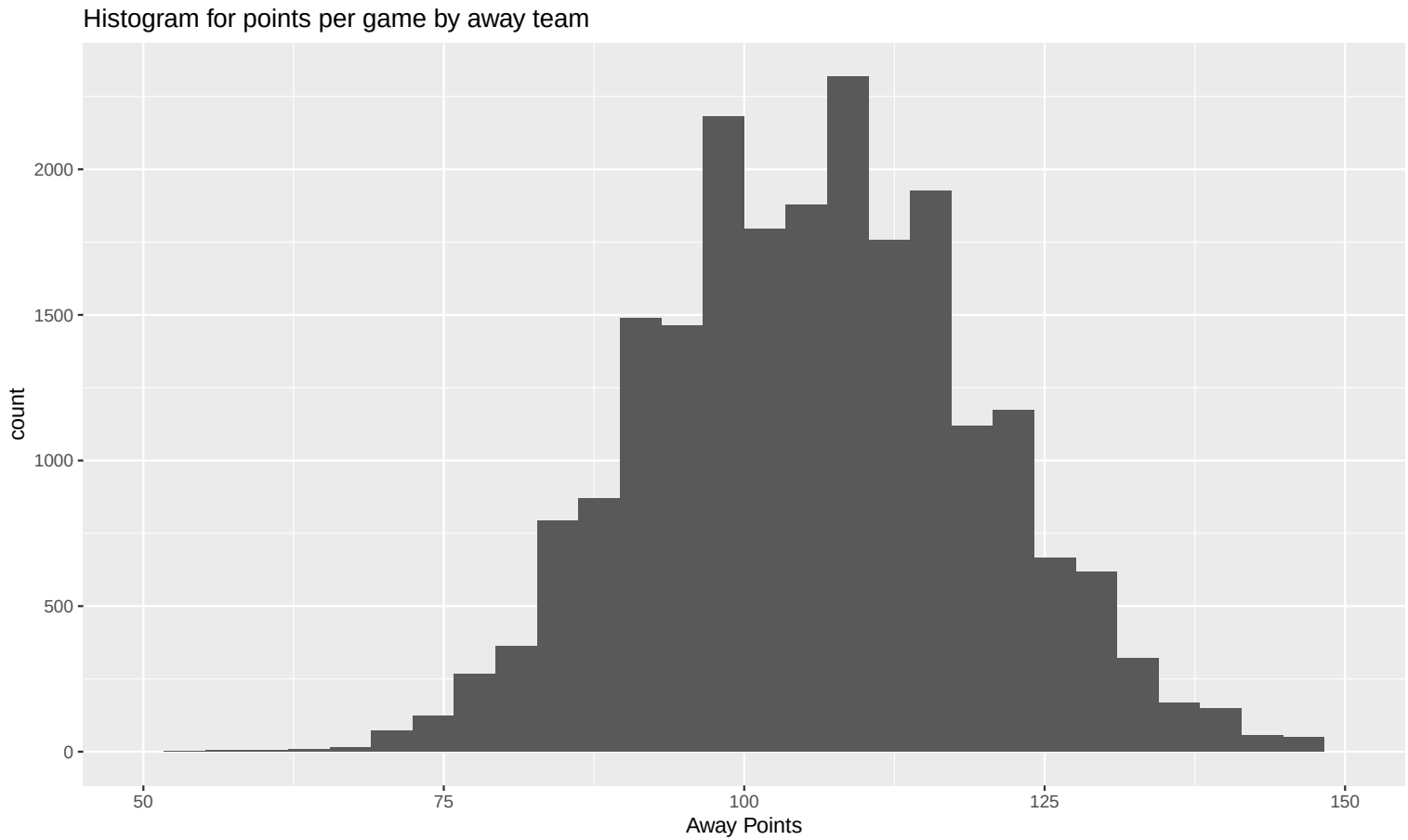
Exclude Boston-Celtics game

```
ggplot(nba_games, aes(x=home.PTS)) + geom_histogram() + xlim(c(50,155))+
  labs(x="Home Points", title="Histogram for points per game by home team")
```



Predicting with distributions

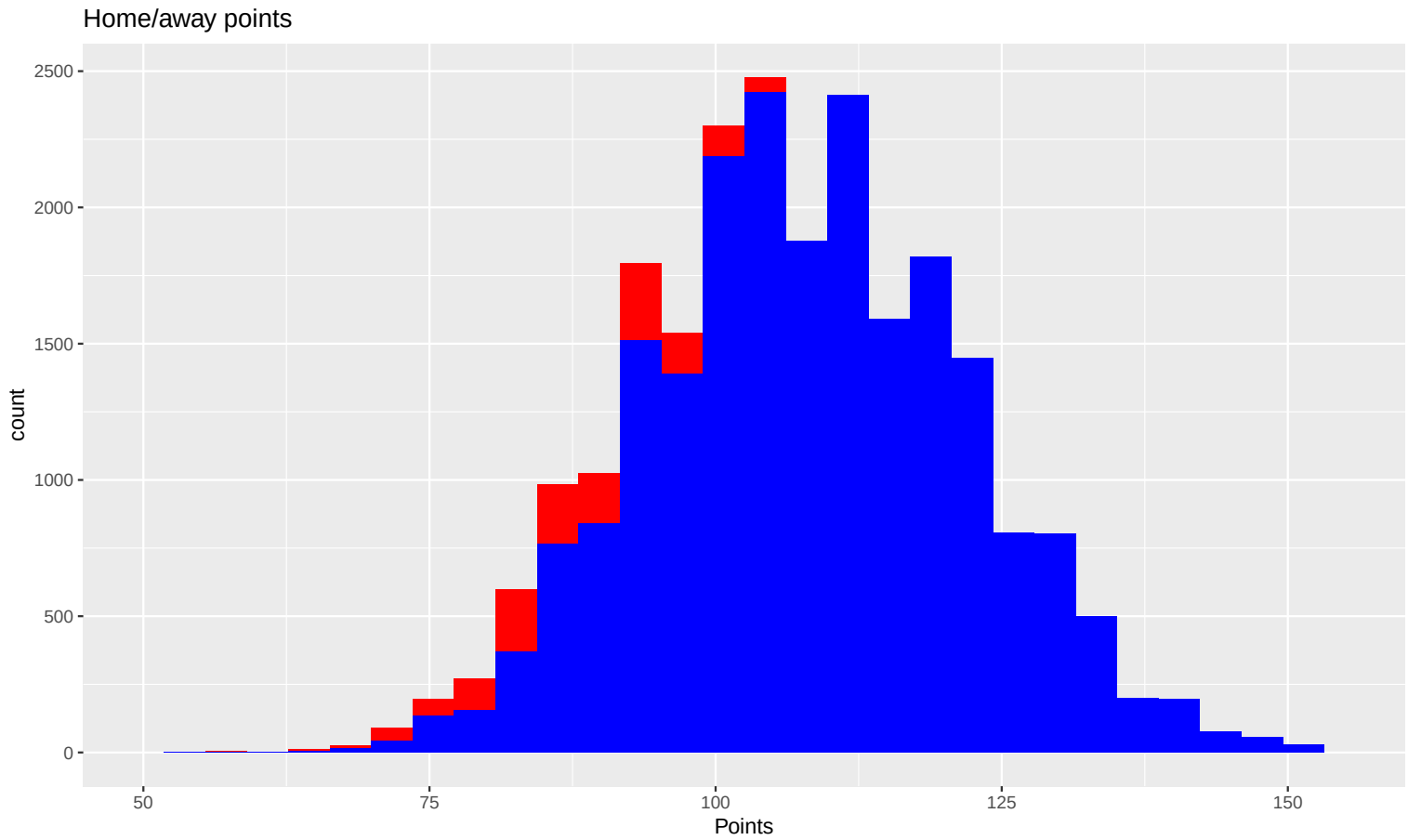
```
ggplot(nba_games, aes(x=away.PTS)) + geom_histogram() + xlim(c(50,150)) +
  labs(x="Away Points", title="Histogram for points per game by away team")
```



Predicting with distributions

Histogram of home and away points together

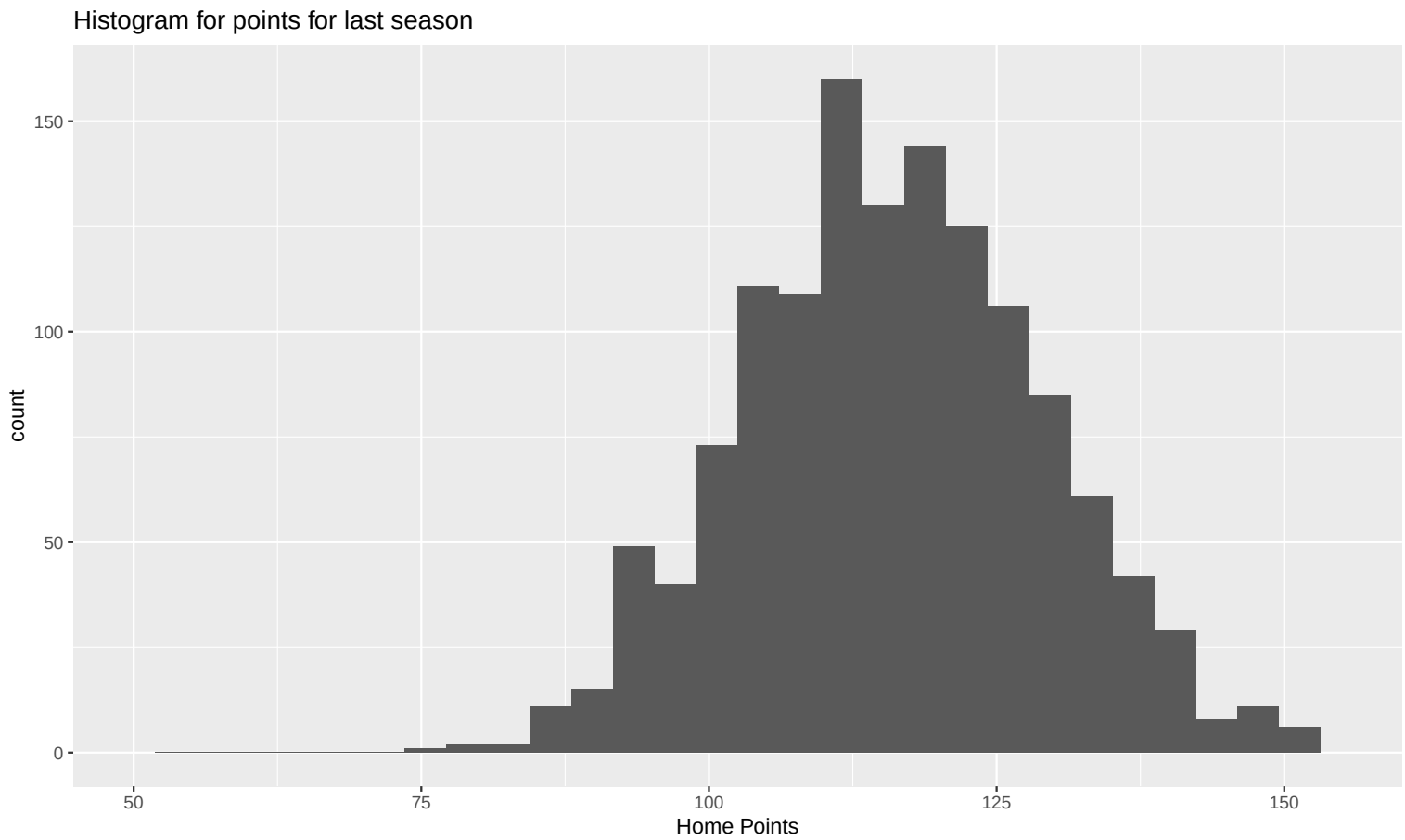
```
ggplot(nba_games) + geom_histogram(aes(x=away.PTS), fill='red') +
  geom_histogram(aes(x=home.PTS),fill='blue') +
  xlim(c(50,155)) + labs(x="Points", title="Home/away points")
```



Predicting with distributions

Home points for last season only

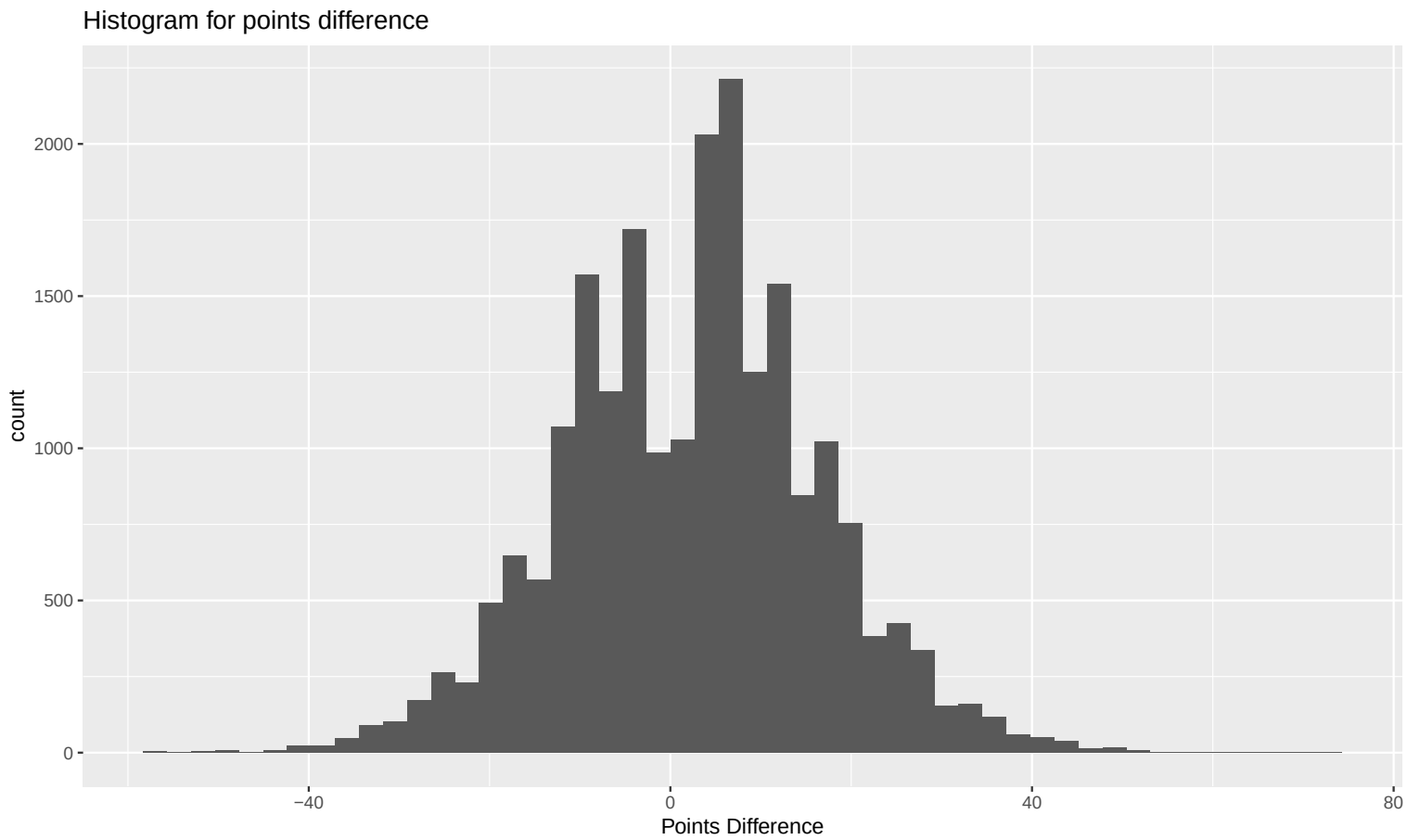
```
nba_games %>% filter(SEASON_ID=='2026') %>%
ggplot(aes(x = home.PTS)) + geom_histogram() + xlim(c(50,155))+
  labs(x="Home Points", title="Histogram for points for last season")
```



Predicting with distributions

What about Points difference ?

```
nba_games$Diff = nba_games$home.PTS-nba_games$away.PTS
ggplot(nba_games, aes(x=Diff)) + geom_histogram(bins=50) +
  labs(x="Points Difference", title="Histogram for points difference")
```



Predicting with distributions

Test hypothesis of Normality:

$$H_0 : \textit{sample came from normal distribution}$$
$$H_1 : \textit{sample did not come from normal distribution}$$

Find the p-value of the test: if α is more than 0.05 then it has a normal distribution failed to reject H_0

Predicting with distributions

```
nba_games %>%
  filter(SEASON_ID=='2026') %>%
  group_by(home.TEAM_ABBREVIATION) %>%
  filter(n() >= 3) %>%  # shapiro.test needs sample size between 3 and 5000
  summarise(p_value = shapiro.test(home.PTS)$p.value)%>%
    arrange(p_value)
## # A tibble: 31 x 2
##    home.TEAM_ABBREVIATION p_value
##    <chr>                    <dbl>
##  1 DEN                     0.0841
##  2 PHX                     0.0924
##  3 NY                      0.116
##  4 SAC                     0.131
##  5 PHI                     0.136
##  6 MIN                     0.143
##  7 BKN                     0.179
##  8 ORL                     0.218
##  9 POR                     0.263
## 10 NO                      0.285
## # i 21 more rows
```

Predicting with distributions

Away points: Normal distribution with $\alpha = 0.05$

```
nba_games %>%
  filter(SEASON_ID=='2026') %>%
  group_by(away.TEAM_ABBREVIATION) %>%
  filter(n() >= 3) %>%  # shapiro.test needs sample size between 3 and 5000
  summarise(p_value = shapiro.test(away.PTS)$p.value)%>%
  arrange(p_value)
## # A tibble: 30 x 2
##    away.TEAM_ABBREVIATION p_value
##    <chr>                    <dbl>
##  1 BKN                     0.0501
##  2 POR                     0.0525
##  3 DEN                     0.0844
##  4 DET                     0.115
##  5 BOS                     0.138
##  6 PHI                     0.144
##  7 PHX                     0.298
##  8 DAL                     0.342
##  9 SAC                     0.407
## 10 ATL                     0.436
## # i 20 more rows
```

Predicting with distributions

Suppose we have two variables with normal distribution,

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- Away team score

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Then the difference between the home and away teams scores will also have normal distribution with the following parameters (assuming this two are independent)

$$\mu_d = \mu_H - m_A$$
$$\sigma_D^2 = \sigma_H^2 + \sigma_A^2$$

Thus the difference will have a normal distribution with the following parameters

$$D \sim N(\mu_H - \mu_A, \sigma_H^2 + \sigma_A^2)$$

Predicting with distributions

Create dataframes with means and standard deviations for each team, for the last season

```
# Exclude All-Star / exhibition sides (too few games for stable variance)
nba_reg = nba_games %>%
  filter(SEASON_ID=='2026',
         !home.TEAM_ABBREVIATION %in% c("STARS","STRIPES","WORLD"),
         !away.TEAM_ABBREVIATION %in% c("STARS","STRIPES","WORLD"))

home = nba_reg %>%
        group_by(home.TEAM_ABBREVIATION) %>%
        summarise(mean = mean(home.PTS),
                  var  = var(home.PTS))

away = nba_reg %>%
        group_by(away.TEAM_ABBREVIATION) %>%
        summarise(mean = mean(away.PTS),
                  var  = var(away.PTS))
```

Predicting with distributions

Dallas Mavericks playing @ Washington Wizards Wizards are the home team, Mavericks are the away team



@



Predicting with distributions

Calculate the mean and standard deviation for the points difference between these two teams

$$s_D = \sqrt{\sigma_H^2 + \sigma_A^2}$$

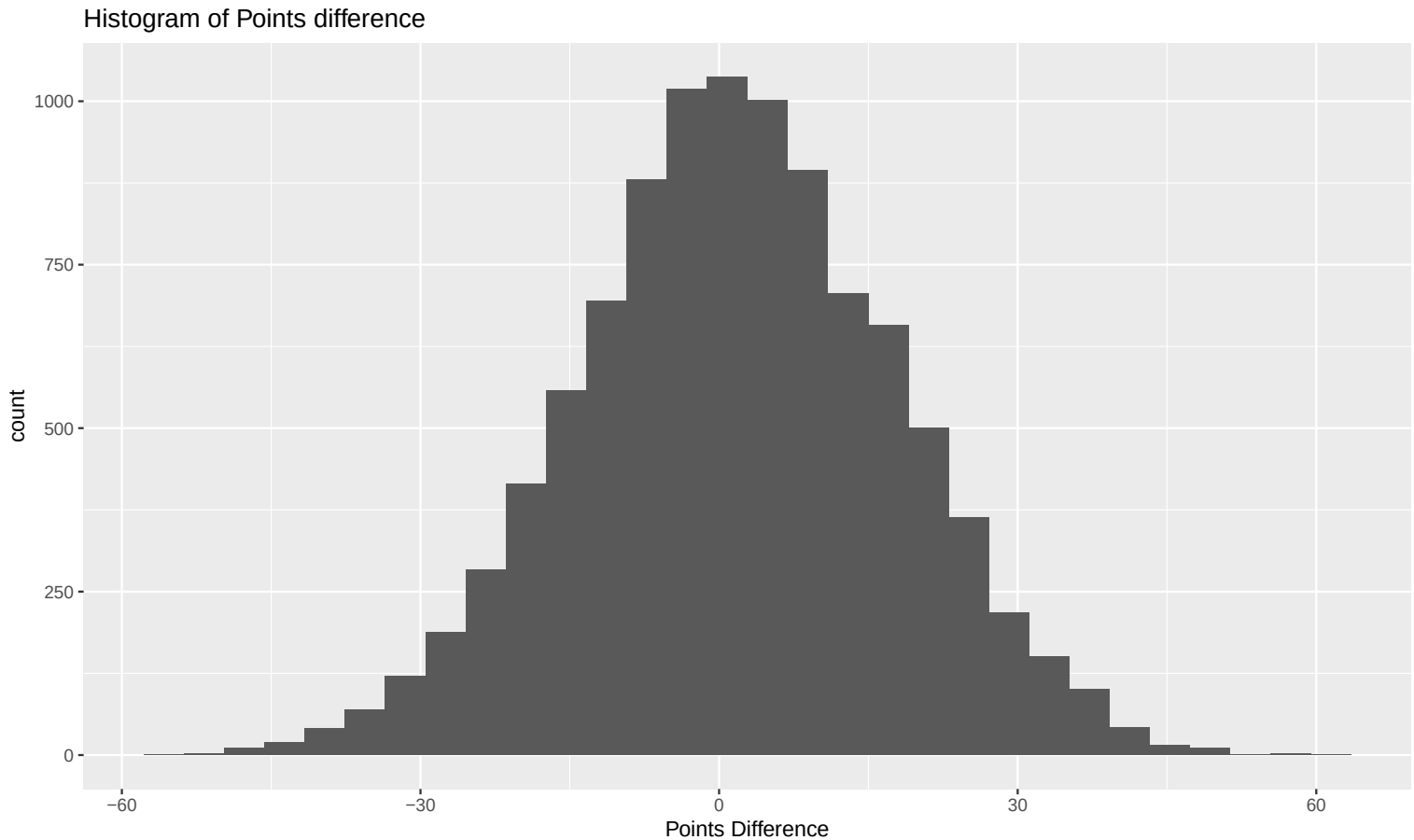
```
(mean = home$mean[home$home.TEAM_ABBREVIATION=="WSH"]-
  away$mean[away$away.TEAM_ABBREVIATION=="DAL"])
## [1] 1.756098

(sd = sqrt(home$var[home$home.TEAM_ABBREVIATION=="WSH"]+
  away$var[away$away.TEAM_ABBREVIATION=="DAL"]))
## [1] 15.66591
```

Predicting with distributions

Simulate 10,000 games and plot the histogram. Can you see the home team advantage ?

```
set.seed(1)
diff = rnorm(10000, mean, sd)
ggplot()+geom_histogram(aes(x=diff))+
  labs(x="Points Difference", title="Histogram of Points difference")
```



Predicting with distributions

Now let's calculate the probabilities of home and away teams to win

- Away team to win $P(x < 0)$
- Home team to win $P(x > 0)$

Predicting with distributions

Probability of away team to win

```
pnorm(0, mean, sd, lower.tail = T)
## [1] 0.4553733
```

Predicting with distributions

Probability of home team to win

```
pnorm(0, mean, sd, lower.tail = F)
## [1] 0.5446267
```

Predicting with distributions

Probability of home team to win with more than 15 points difference

Home team will win with spread of 15 points

```
pnorm(15, mean, sd, lower.tail = F)
## [1] 0.1989448
```

Predicting with distributions

```
home_away = list(home,away)
probs = function(data, home, away, spread=0){
  home_df = data[[1]]
  away_df = data[[2]]
  mean_home = home_df$mean[home_df$home.TEAM_ABBREVIATION==home]
  mean_away = away_df$mean[away_df$away.TEAM_ABBREVIATION==away]
  mean_diff = mean_home-mean_away
  var_home = home_df$var[home_df$home.TEAM_ABBREVIATION==home]
  var_away = away_df$var[away_df$away.TEAM_ABBREVIATION==away]
  sd_diff = sqrt(var_home+var_away)
  df = data.frame(pnorm(spread, mean_diff, sd_diff, lower.tail = F),
                  pnorm(spread, mean_diff, sd_diff))
  colnames(df) = c(home, away)
  return(df)
}
```

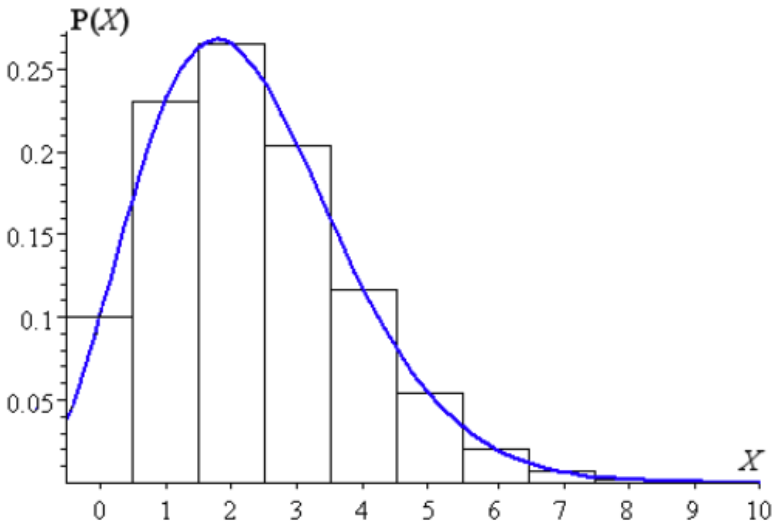
Predicting with distributions

```
probs(home_away, home="WSH", away="DAL", 0)
##         WSH       DAL
## 1 0.5446267 0.4553733
```

Predicting football games results

Predicting football games results

Let's recall what is Poisson distribution about



$$\lambda = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$f(x;\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Predicting football games results

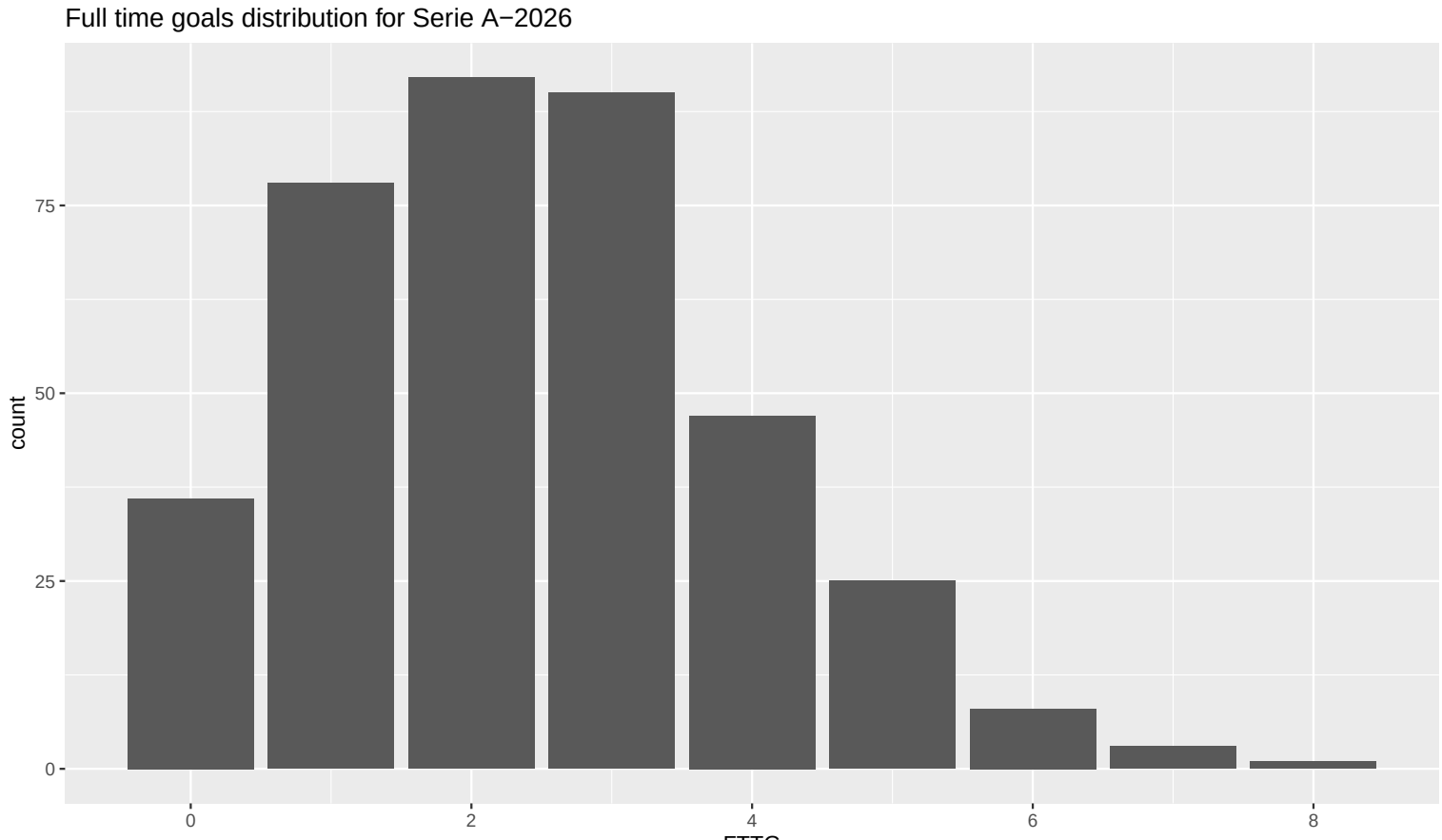
Dataset

```
data(f_data_sm)
head(f_data_sm)
##   SEASON COUNTRY          LEAGUE       DATE    HOMETEAM
## 1   1994 England Premier League 1993-08-14     Arsenal
## 2   1994 England Premier League 1993-08-14 Aston Villa
## 3   1994 England Premier League 1993-08-14     Chelsea
## 4   1994 England Premier League 1993-08-14   Liverpool
## 5   1994 England Premier League 1993-08-14    Man City
## 6   1994 England Premier League 1993-08-14   Newcastle
##          AWAYTEAM FTSC FTR FTHG FTAG FTTG  H  D  A
## 1        Coventry  0-3   A    0    3    3 NA NA NA
## 2             QPR  4-1   H    4    1    5 NA NA NA
## 3       Blackburn  1-2   A    1    2    3 NA NA NA
## 4 Sheffield Weds  2-0   H    2    0    2 NA NA NA
## 5           Leeds  1-1   D    1    1    2 NA NA NA
## 6       Tottenham  0-1   A    0    1    1 NA NA NA
```

Predicting football games results

Full time total goals distribution for Serie A, Season 2025-2026

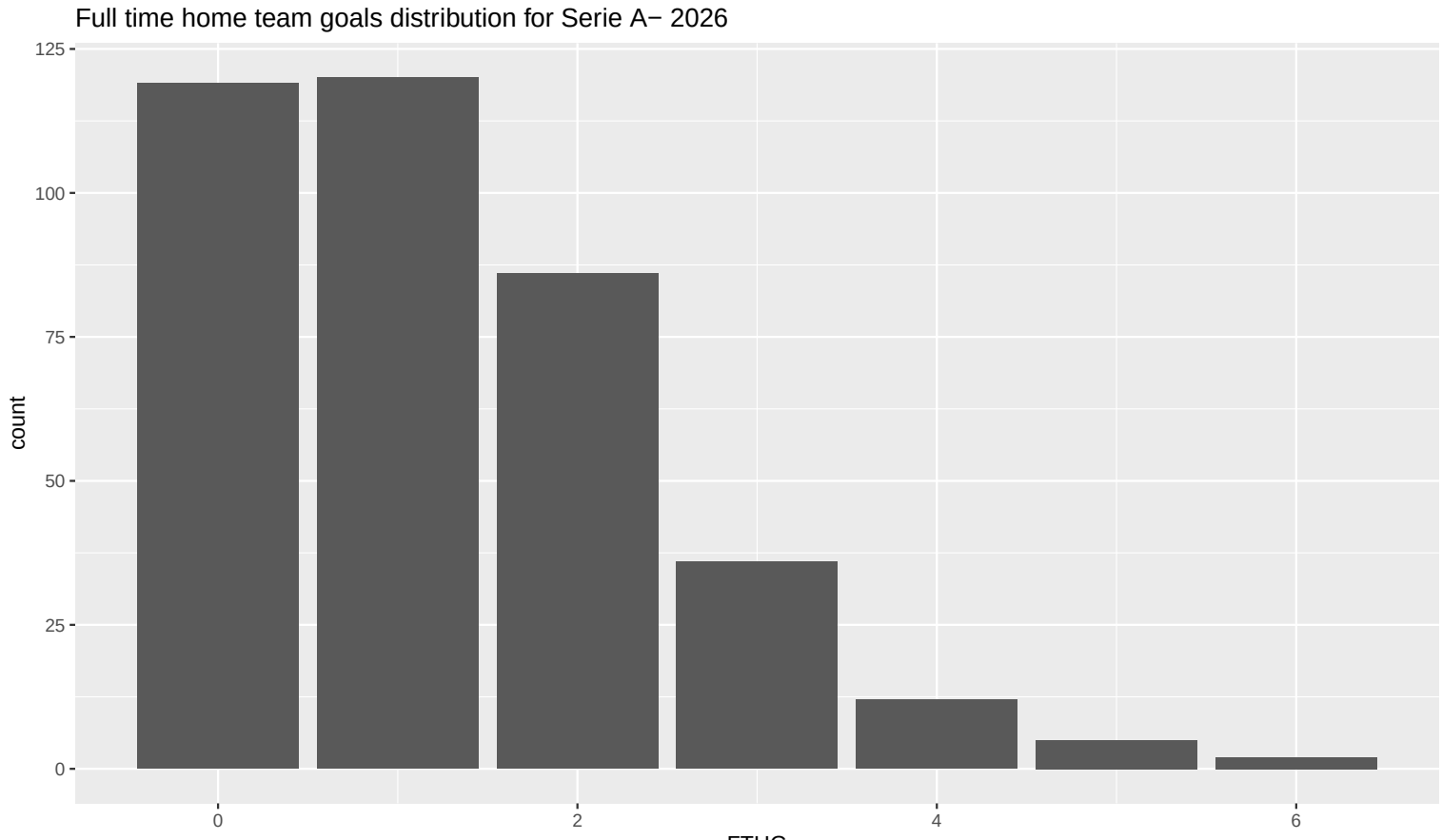
```
f_data_sm %>%
  filter(COUNTRY=="Italy", SEASON=="2026") %>%
ggplot(aes(x=FTTG))+geom_bar()+
ggtitle("Full time goals distribution for Serie A-2026")
```



Predicting football games results

Distribution of home team full time goals

```
f_data_sm %>%
  filter(COUNTRY=="Italy", SEASON=="2026") %>%
ggplot(aes(x=FTHG))+geom_bar()+
ggtitle("Full time home team goals distribution for Serie A- 2026")
```



Predicting football games results

Does the total number of goals scored follows Poisson distribution?

First find the λ (average number of goals per game)

```
f_data_sm %>%
  filter(COUNTRY=="Italy", SEASON=='2026') %>%
  summarise(mean(FTTG))
##   mean(FTTG)
## 1   2.426316
```

Predicting football games results

We can test the hypothesis, whether the goal's distribution is Poisson: Our hypothesis is the following:

$$H_0 : \textit{The variables have Poisson distribution}$$

$$H_1 : \textit{The variables do not have Poisson distribution}$$

We will test the hypothesis with Chi-square test: thus comparing empirical probability distribution (from the data) and hypothethical Poisson distribution

Predicting football games results

Goal frequencies

```
(seriea = f_data_sm %>%
          filter(COUNTRY=="Italy", SEASON=="2026") %>%
          group_by(FTHG) %>%
          summarise(count = n()) %>%
          as.data.frame())
##   FTHG count
## 1    0   119
## 2    1   120
## 3    2    86
## 4    3    36
## 5    4    12
## 6    5     5
## 7    6     2
```

Predicting football games results

We will use chi-square test

First create a vector with observed relative frequencies

```
(rel_fr=seriea$count/sum(seriea$count))
## [1] 0.313157895 0.315789474 0.226315789 0.094736842 0.031578947
## [6] 0.013157895 0.005263158
```

```
(h_goals_mean_italy <- (f_data_sm %>%
                        filter(COUNTRY=="Italy", SEASON=="2026") %>%
                          summarize(mean(FTHG)))$`mean(FTHG)`)
## [1] 1.276316
```

Then create expected relative frequencies (probabilities), under the assumption that the variable follows Poisson distribution with λ=1.2763158

```
(probs=dpois(c(0:6), lambda=h_goals_mean_italy))
## [1] 0.279063538 0.356173199 0.227294739 0.096699955 0.030854920
## [6] 0.007876124 0.001675404
```

Predicting football games results

Is there a significant difference between these two ?

```
chisq.test(rel_fr, p=probs, rescale.p = T, simulate.p.value = TRUE)
##
##  Chi-squared test for given probabilities with simulated
##  p-value (based on 2000 replicates)
##
## data:  rel_fr
## X-squared = 0.020023, df = NA, p-value = 1
```

High p-value talks in favor for NULL hypothesis, meaning that the observed result is unlikely to happen by chance. We failed to reject the null hypothesis, thus we have a Poisson distribution

Predicting football games results

Lazio (Home team) vs. Fiorentina(Away team)



VS



Predicting football games results

- We assume that probabilities for Lazio and Fiorentina scoring are independent from each other.
- Thus: $P(L = 0 \cap F = 0) = P(L = 0) * P(F = 0)...$
- If you want to find the probability of the draw then, $P(L = 0 \cap F = 0 \cup L = 1 \cap F = 1 \cup L = 2 \cap F = 2)$
- As $\{L = 0 \cap F = 0 \cup L = 1 \cap F = 1 \cup L = 2 \cap F = 2\}$ events are mutually exclusive, the probability of the draw can be calculated as.

$$P(L = 0 \cap F = 0) + P(L = 1 \cap F = 1) + P(L = 2 \cap F = 2)... =$$
$$P(L = 0) * P(F = 0) + P(L = 1) * P(F = 1)...$$

Predicting football games results

Means (λ parameters) for home and away teams

```
data("f_data_sm")

home_means = f_data_sm %>%
                filter(COUNTRY=="Italy", SEASON=="2026") %>%
                group_by(HOMETEAM) %>%
                summarise(mean = mean(FTHG))

away_means = f_data_sm %>%
                filter(COUNTRY=="Italy", SEASON=="2026") %>%
                group_by(AWAYTEAM) %>%
                summarise(mean = mean(FTAG))
```

Predicting football games results

For Lazio as a home team and for Fiorentina as an away team

```
(Lazio = home_means$mean[home_means$HOMETEAM=="Lazio"])
## [1] 1.421053

(Fiorentina= away_means$mean[away_means$AWAYTEAM=="Fiorentina"])
## [1] 1.052632
```

Predicting football games results

Generate vector of probabilities for number of goals scored for both teams.

Range from 0 to 9

```
options(scipen=999)
goal_probs_lazio=dpois(c(0:9), lambda=Lazio)
goal_probs_fiorentina=dpois(c(0:9), lambda=Fiorentina)
```

```
goal_probs_lazio
##  [1] 0.24145971496 0.34312696336 0.24380073712 0.11548455969
##  [5] 0.04102740936 0.01166042161 0.00276167880 0.00056064156
##  [9] 0.00009958765 0.00001572437
```

```
goal_probs_fiorentina
##  [1] 0.349018070931 0.367387443086 0.193361812150 0.067846249877
##  [5] 0.017854276284 0.003758795007 0.000659437721 0.000099163567
##  [9] 0.000013047838 0.000001526063
```

Predicting football games results

Multiplying two vectors of probabilities we will get the matrix of probabilities for a given result

```
options(width = 300)
a=goal_probs_fiorentina %*% t(goal_probs_lazio)
a=round(a, digits=4)
print(a)
##          [,1]   [,2]   [,3]   [,4]   [,5]   [,6]   [,7]   [,8] [,9] [,10]
##  [1,] 0.0843 0.1198 0.0851 0.0403 0.0143 0.0041 0.0010 0.0002    0     0
##  [2,] 0.0887 0.1261 0.0896 0.0424 0.0151 0.0043 0.0010 0.0002    0     0
##  [3,] 0.0467 0.0663 0.0471 0.0223 0.0079 0.0023 0.0005 0.0001    0     0
##  [4,] 0.0164 0.0233 0.0165 0.0078 0.0028 0.0008 0.0002 0.0000    0     0
##  [5,] 0.0043 0.0061 0.0044 0.0021 0.0007 0.0002 0.0000 0.0000    0     0
##  [6,] 0.0009 0.0013 0.0009 0.0004 0.0002 0.0000 0.0000 0.0000    0     0
##  [7,] 0.0002 0.0002 0.0002 0.0001 0.0000 0.0000 0.0000 0.0000    0     0
##  [8,] 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000    0     0
##  [9,] 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000    0     0
## [10,] 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000    0     0
```

Predicting football games results

The probability that the game will end 0-0

	0	1	2	3	4	5	6	7	8	9
0	**0.0843**	0.1198	0.0851	0.0403	0.0143	0.0041	0.0010	0.0002	0	0
1	0.0887	0.1261	0.0896	0.0424	0.0151	0.0043	0.0010	0.0002	0	0
2	0.0467	0.0663	0.0471	0.0223	0.0079	0.0023	0.0005	0.0001	0	0
3	0.0164	0.0233	0.0165	0.0078	0.0028	0.0008	0.0002	0.0000	0	0
4	0.0043	0.0061	0.0044	0.0021	0.0007	0.0002	0.0000	0.0000	0	0
5	0.0009	0.0013	0.0009	0.0004	0.0002	0.0000	0.0000	0.0000	0	0
6	0.0002	0.0002	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0	0
7	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0	0
8	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0	0
9	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0	0

Predicting football games results

3 1 Lazio winning

	0	1	2	3	4	5	6	7	8	9
0	0.0843	0.1198	0.0851	0.0403	0.0143	0.0041	0.0010	0.0002	0	0
1	0.0887	0.1261	0.0896	0.0424	0.0151	0.0043	0.0010	0.0002	0	0
2	0.0467	0.0663	0.0471	0.0223	0.0079	0.0023	0.0005	0.0001	0	0
3	0.0164	**0.0233**	0.0165	0.0078	0.0028	0.0008	0.0002	0.0000	0	0
4	0.0043	0.0061	0.0044	0.0021	0.0007	0.0002	0.0000	0.0000	0	0
5	0.0009	0.0013	0.0009	0.0004	0.0002	0.0000	0.0000	0.0000	0	0
6	0.0002	0.0002	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0	0
7	0.0000	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0	0
8	0.0000	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0	0
9	0.0000	0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0	0

Predicting football games results

Probability of draw: sum all elements on the diagonal

	0	1	2	3	4	5	6	7	8	9
0	**0.0843**	0.1198	0.0851	0.0403	0.0143	0.0041	0.001	0.0002	0	0
1	0.0887	**0.1261**	0.0896	0.0424	0.0151	0.0043	0.001	0.0002	0	0
2	0.0467	0.0663	**0.0471**	0.0223	0.0079	0.0023	0.0005	0.0001	0	0
3	0.0164	0.0233	0.0165	**0.0078**	0.0028	0.0008	0.0002	0	0	0
4	0.0043	0.0061	0.0044	0.0021	**0.0007**	0.0002	0	0	0	0
5	0.0009	0.0013	0.0009	0.0004	0.0002	**0**	0	0	0	0
6	0.0002	0.0002	0.0002	0.0001	0	0	**0**	0	0	0
7	0	0	0	0	0	0	0	**0**	0	0
8	0	0	0	0	0	0	0	0	**0**	0
9	0	0	0	0	0	0	0	0	0	**0**

Predicting football games results

Probability of Fiorentina winning: Sum of all the probabilities in the lower triangle, without diagonal

	0	1	2	3	4	5	6	7	8	9
0	0.0843	0.1198	0.0851	0.0403	0.0143	0.0041	0.001	0.0002	0	0
1	**0.0887**	0.1261	0.0896	0.0424	0.0151	0.0043	0.001	0.0002	0	0
2	**0.0467**	**0.0663**	0.0471	0.0223	0.0079	0.0023	0.0005	0.0001	0	0
3	**0.0164**	**0.0233**	**0.0165**	0.0078	0.0028	0.0008	0.0002	0	0	0
4	**0.0043**	**0.0061**	**0.0044**	**0.0021**	0.0007	0.0002	0	0	0	0
5	**0.0009**	**0.0013**	**0.0009**	**0.0004**	**0.0002**	0	0	0	0	0
6	**0.0002**	**0.0002**	**0.0002**	**0.0001**	**0**	**0**	0	0	0	0
7	**0**	**0**	**0**	**0**	**0**	**0**	**0**	0	0	0
8	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	0	0
9	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	0

Predicting football games results

Probability of Lazio winning: Sum of all the probabilities in the upper triangle, without diagonal

	0	1	2	3	4	5	6	7	8	9
0	0.0843	**0.1198**	**0.0851**	**0.0403**	**0.0143**	**0.0041**	**0.001**	**0.0002**	**0**	**0**
1	0.0887	0.1261	**0.0896**	**0.0424**	**0.0151**	**0.0043**	**0.001**	**0.0002**	**0**	**0**
2	0.0467	0.0663	0.0471	**0.0223**	**0.0079**	**0.0023**	**0.0005**	**0.0001**	**0**	**0**
3	0.0164	0.0233	0.0165	0.0078	**0.0028**	**0.0008**	**0.0002**	**0**	**0**	**0**
4	0.0043	0.0061	0.0044	0.0021	0.0007	**0.0002**	**0**	**0**	**0**	**0**
5	0.0009	0.0013	0.0009	0.0004	0.0002	0	**0**	**0**	**0**	**0**
6	0.0002	0.0002	0.0002	0.0001	0	0	0	**0**	**0**	**0**
7	0.0000	0	0	0	0	0	0	0	**0**	**0**
8	0.0000	0	0	0	0	0	0	0	0	**0**
9	0.0000	0	0	0	0	0	0	0	0	0

Predicting football games results

Probability of game ending in draw

```
sum(diag(a))
## [1] 0.266
```

Probability of Fiorentina winning

```
sum(a[lower.tri(a, diag=F)])
## [1] 0.2792
```

Probability of Lazio winning

```
sum(a[upper.tri(a, diag=F)])
## [1] 0.4545
```

Predicting football games results

- Suppose X is a Poisson random variable with mean λ_x that describes the number of the goals the home team scores on average.
- Suppose Y is a Poisson random variable with mean λ_y that describes the number of the goals the away team scores on average.
- $Z = X - Y$ random variable has a Skellam distribution and Z represents the outcome of the game with respect to the Home team. If $Z > 0$, then the home team has won, if $Z = 0$, then it was a draw, and if $Z < 0$, the home team has lost.

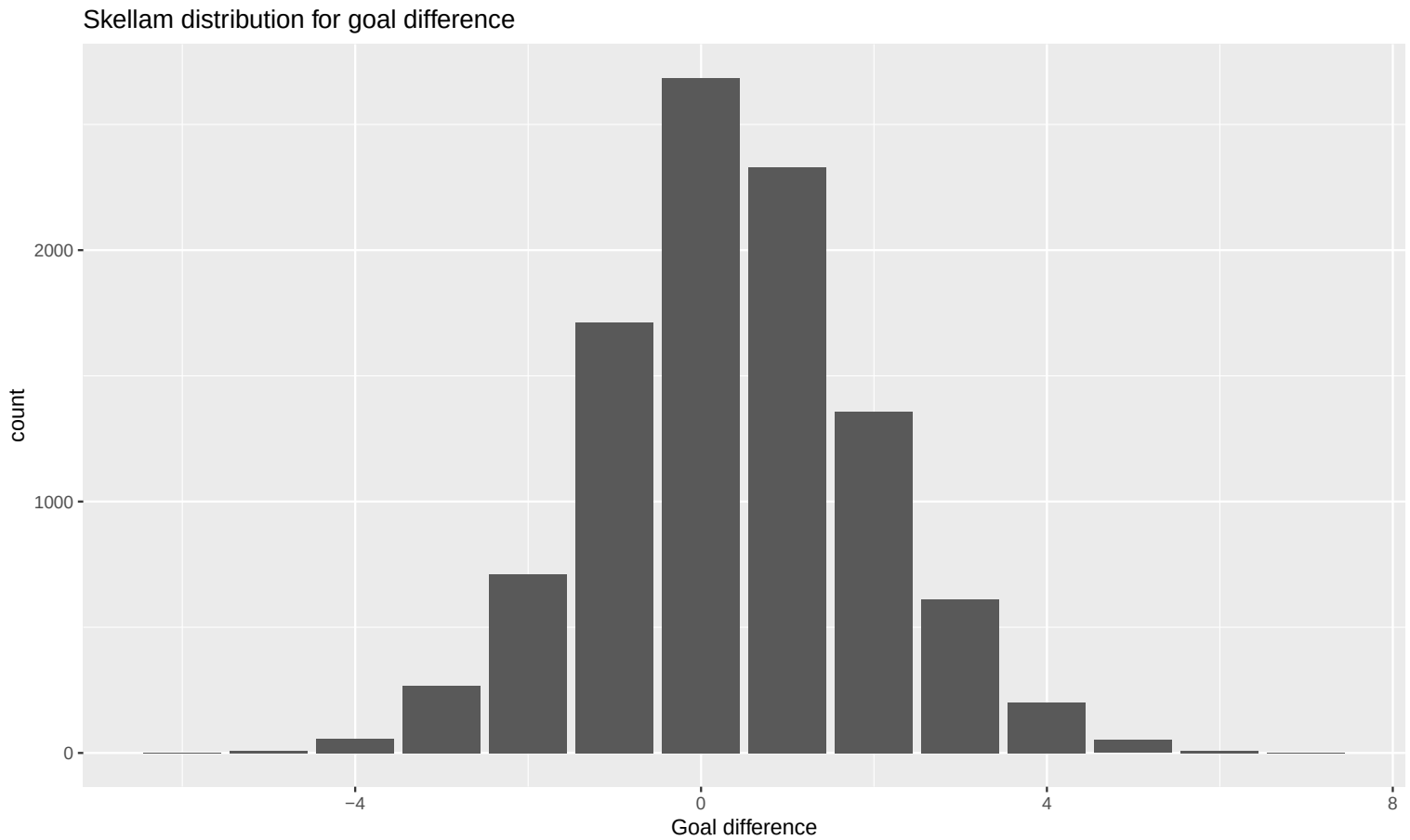
Library for skellam distribution

```
if (!requireNamespace("VGAM", quietly = TRUE)) install.packages("VGAM")
library(VGAM)
```

Predicting football games results

Histogram for goal difference

```
set.seed(1)
k = rskellam(10000,mu1=Lazio, mu2=Fiorentina)
ggplot()+geom_bar(aes(x=k))+
  labs(x="Goal difference", title="Skellam distribution for goal difference")
```



Predicting football games results

Calculating probabilities

Probability of home team to win

```
sum(k>0)/10000
## [1] 0.4558
```

Probability of draw

```
sum(k==0)/10000
## [1] 0.2685
```

Probability of away team to win

```
sum(k<0)/10000
## [1] 0.2757
```

Predicting with Ratings

Predicting with Ratings

- Every team can have some kind of rating, describing its ability to win.
- In football (soccer) the rating can be expressed as the number of points earned, number of goals scored, number of goals conceded, etc.
- One way to go with rating football teams is to calculate goal supremacy for each team.

Goal supremacy = Number of scored goals- Number of conceded goals

Predicting with Ratings

- Tottenham playing Leeds United (2002).
- Looking at the goal supremacy (goal difference) for each team for their last 6 games
- Both teams will have Goal supremacy rating (we will call it Power Rating) of -3
- What will be your best guess for probability of Tottenham to win? Draw ?

Match	Tottenham	Leeds
1	Arsenal 3:0 Tottenham	Leeds 2:4 Bolton
2	Sunderland 2:0 Tottenham	West Ham 3:4 Leeds
3	Tottenham 0:0 Chelsea	Leeds 0:1 Everton
4	Liverpool 2:1 Tottenham	Middlesbrough 2:2 Leeds
5	Tottenham 3:1 Bolton	Leeds 0:1 Liverpool
6	Blackburn 1:2 Tottenham	Aston Villa 0:0 Leeds

Adapted from: "Fixed Odds Sports Betting" – Joseph Buchdahl

Predicting with Ratings

Steps for Power Rating calculation

- Get the Goal Supremacy Rating (Power) for each team n games before the given game.
- Calculate Power Difference for home/away teams – Match rating
- Look at the relative frequencies for home team win, draw and away team win for the given power difference.
- Those will be the probabilities

Predicting with Ratings

Kind of table we need

Match rating	No. of home wins	No. of draws	No. of away wins	% of total matches	% of home wins	% of home draws	% of away wins
-11	75	54	91	1.57%	34.1%	24.5%	41.4%
-10	84	94	91	1.92%	31.2%	34.9%	33.8%
-9	123	91	112	2.33%	37.7%	27.9%	34.4%
-8	171	113	124	2.91%	41.9%	27.7%	30.4%
-7	190	121	170	3.44%	39.5%	25.2%	35.3%
-6	242	202	191	4.54%	38.1%	31.8%	30.1%
-5	279	212	197	4.91%	40.6%	30.8%	28.6%
-4	293	219	215	5.19%	40.3%	30.1%	29.6%

Predicting with Ratings

- Subset the f_data_sm to get only the games for England
- Select columns with team names, date, season indicator and final score

```
data(f_data_sm)
england = f_data_sm %>%
  filter(COUNTRY == "England") %>%
  arrange(DATE) %>%
  select("SEASON","DATE","HOMETEAM","AWAYTEAM","FTHG", "FTAG")%>%
  mutate(ID = 1:n())
```

Predicting with Ratings

Reshape to long format: one row per team in each game

```
# Vectorized last-6 power ratings (same idea as a game-by-game loop, much faster)
england_long <- reshape2::melt(
  england,
  id.vars = c("ID", "HOMETEAM", "AWAYTEAM", "SEASON", "DATE"),
  measure.vars = c("FTHG", "FTAG"),
  variable.name = "H_A"
) %>%
  arrange(ID) %>%
  mutate(
    TEAM1 = ifelse(H_A == "FTAG", AWAYTEAM, HOMETEAM),
    TEAM2 = ifelse(H_A == "FTAG", HOMETEAM, AWAYTEAM)
  ) %>%
  select(ID, SEASON, DATE, TEAM1, TEAM2, H_A, value)
```

Predicting with Ratings

For each team, goals scored and conceded in the last 6 games of the same season

```
england_long1 <- england_long %>%
  group_by(SEASON, TEAM1) %>%
  mutate(CUM_SC = dplyr::lag(
    zoo::rollapply(value, 6, sum, fill = NA, align = "right"), 1)) %>%
  ungroup() %>%
  group_by(SEASON, TEAM2) %>%
  mutate(CUM_C = dplyr::lag(
    zoo::rollapply(value, 6, sum, fill = NA, align = "right"), 1)) %>%
  ungroup() %>%
  filter(!is.na(CUM_SC), !is.na(CUM_C)) %>%
  select(ID, H_A, value, CUM_SC, CUM_C) %>%
  as.data.frame()
```

Predicting with Ratings

Reshape back to wide format: one row per game again

```
england_wide <- reshape(
  england_long1,
  direction = "wide",
  timevar = "H_A",
  idvar = "ID",
  v.names = c("value", "CUM_SC", "CUM_C")
)
```

Predicting with Ratings

Calculate POWERHOME, POWERAWAY, POWERDIFF and RESULT

```
england <- england %>%
  inner_join(england_wide, by = "ID") %>%
  mutate(
    POWERHOME = CUM_SC.FTHG - CUM_C.FTAG,
    POWERAWAY = CUM_SC.FTAG - CUM_C.FTHG,
    POWERDIFF = POWERHOME - POWERAWAY,
    RESULT = ifelse(FTHG > FTAG, "HW",
                    ifelse(FTHG < FTAG, "AW", "D"))
  ) %>%
  select(SEASON, DATE, HOMETEAM, AWAYTEAM, FTHG, FTAG,
         POWERHOME, POWERAWAY, POWERDIFF, RESULT) %>%
  filter(!is.na(POWERDIFF))
```

Predicting with Ratings

Take out remaining NA's

```
options(width=80)
england = england[! is.na(england$POWERDIFF),]
head(england)
##   SEASON       DATE    HOMETEAM   AWAYTEAM FTHG FTAG POWERHOME POWERAWAY
## 1   1994 1993-09-11     Arsenal    Ipswich    4    0         2         4
## 2   1994 1993-09-11 Aston Villa   Coventry    0    0         4         5
## 3   1994 1993-09-11     Chelsea Man United    1    0         0        11
## 4   1994 1993-09-11    Man City        QPR    3    0        -2        -1
## 5   1994 1993-09-11     Norwich  Wimbledon    0    1         4         2
## 6   1994 1993-09-11      Oldham    Everton    0    1        -4         1
##   POWERDIFF RESULT
## 1        -2     HW
## 2        -1      D
## 3       -11     HW
## 4        -1     HW
## 5         2     AW
## 6        -5     AW
```

Predicting with Ratings

Now let's calculate the game result frequencies by Match rating (POWERDIFF)

```
freqs = dcast(england, POWERDIFF~RESULT)
freqs[is.na(freqs)] <- 0
head(freqs)
##   POWERDIFF AW D HW
## 1       -32  1 0  0
## 2       -30  1 0  0
## 3       -29  0 0  1
## 4       -26  4 0  0
## 5       -25  2 0  0
## 6       -24  3 0  0
```

Predicting with Ratings

Add relative frequencies

```
freqs = data.frame(freqs, freqs[,2:4]/rowSums(freqs[,2:4]))
head(freqs)
##   POWERDIFF AW D HW AW.1 D.1 HW.1
## 1       -32  1 0  0    1   0    0
## 2       -30  1 0  0    1   0    0
## 3       -29  0 0  1    0   0    1
## 4       -26  4 0  0    1   0    0
## 5       -25  2 0  0    1   0    0
## 6       -24  3 0  0    1   0    0
```

Predicting with Ratings

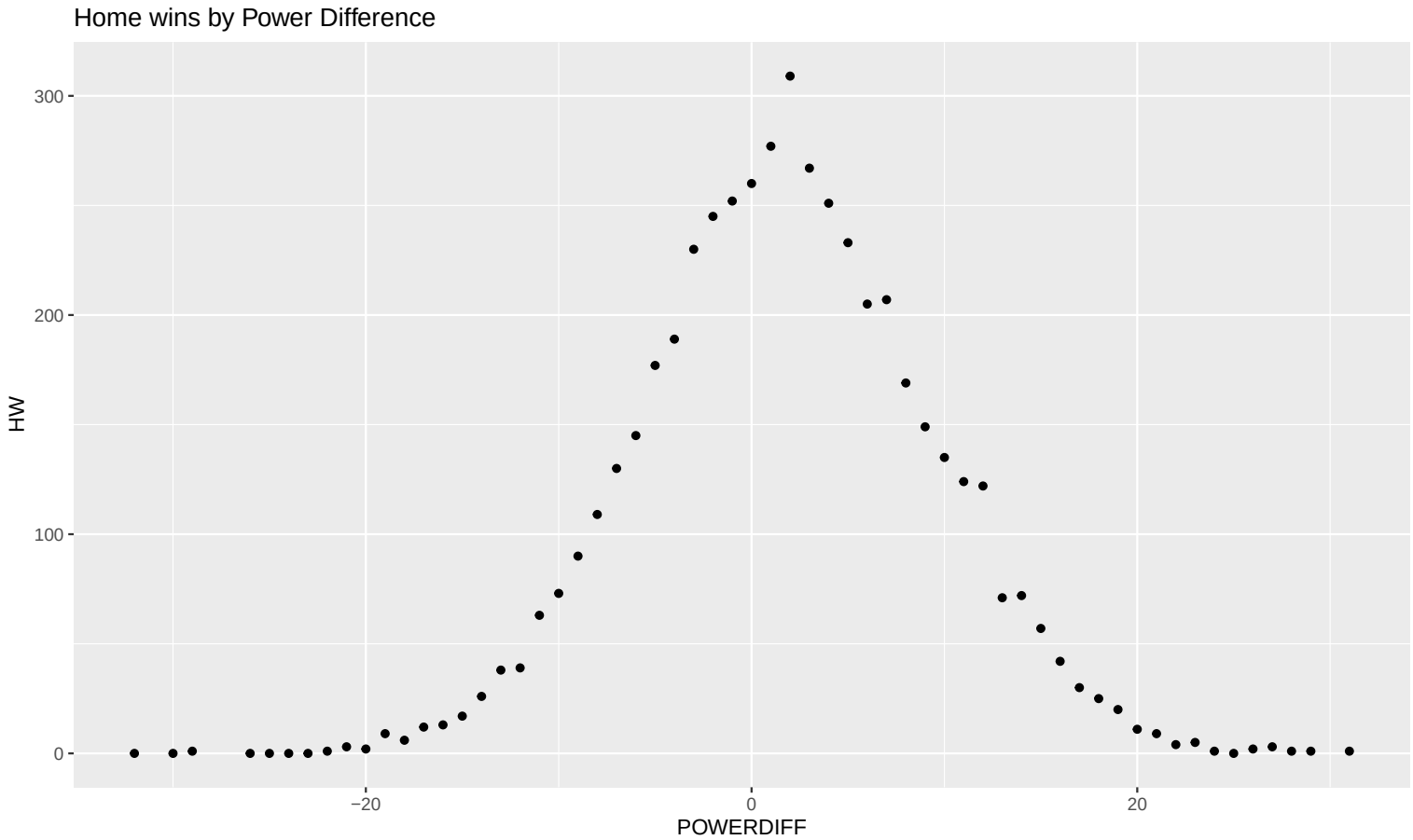
Tottenham vs. Leeds United, two teams with equal Goal Supremacy, the POWERDIFF=0

```
freqs[freqs$POWERDIFF==0,]
##    POWERDIFF  AW   D  HW     AW.1      D.1      HW.1
## 30         0 147 172 260 0.253886 0.2970639 0.4490501
```

Predicting with Ratings

Distribution of Home Wins by Power difference

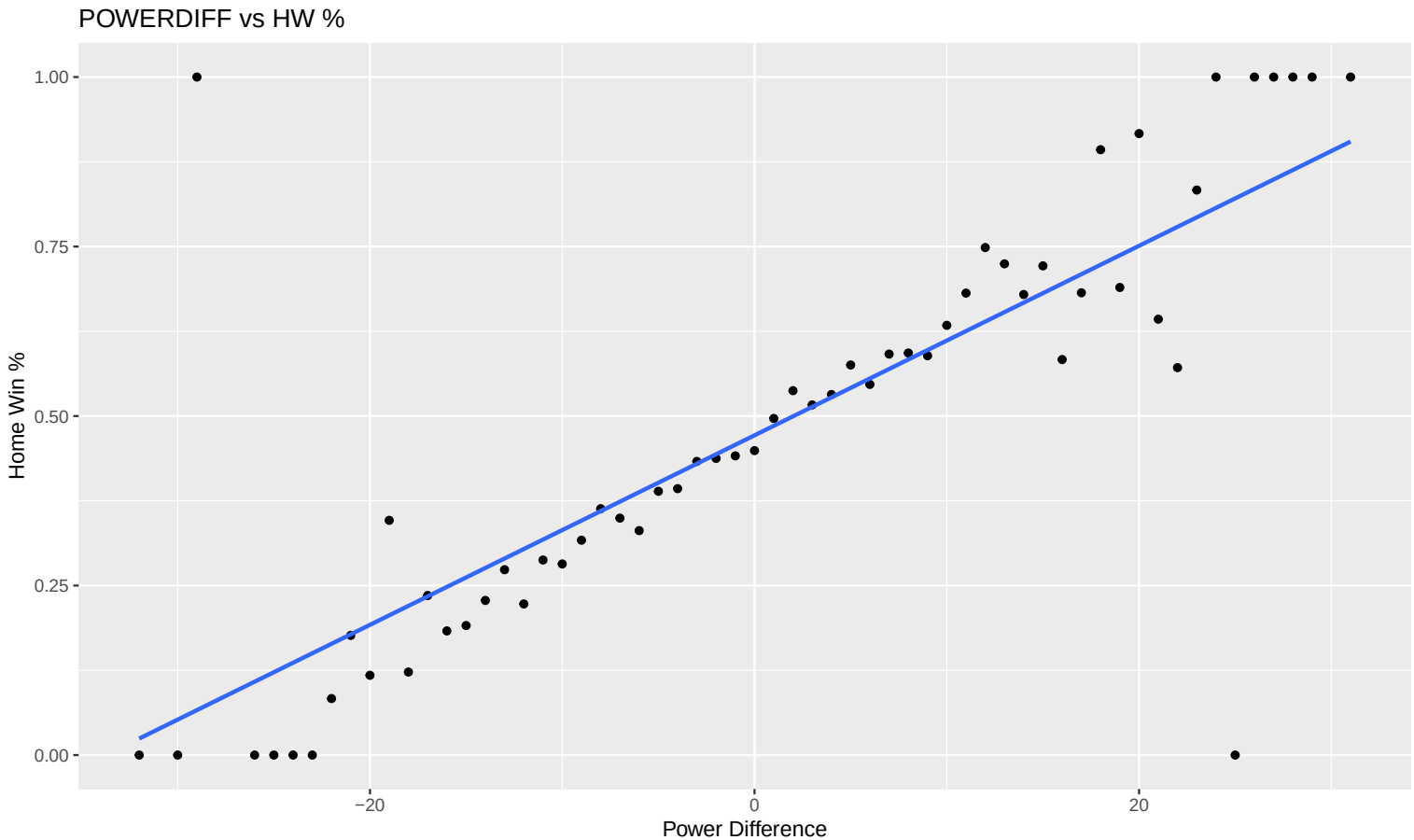
```
ggplot(freqs, aes(x=POWERDIFF, y = HW))+geom_point()+
  ggtitle("Home wins by Power Difference")
```



Predicting with Ratings

Look at the relationship between probability of home team to win and power difference, linear with exception of some outliers

```
ggplot(freqs, aes(x=POWERDIFF, y=HW.1))+geom_point()+
  geom_smooth(method="lm", se=F)+
  labs(x="Power Difference", y="Home Win %", title = "POWERDIFF vs HW %")
```



Predicting with Ratings

Look at the model

```
modelH = lm(HW.1~POWERDIFF, data=freqs)
summary(modelH)
##
## Call:
## lm(formula = HW.1 ~ POWERDIFF, data = freqs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.82100 -0.05340 -0.00862  0.03845  0.93372
##
## Coefficients:
##             Estimate Std. Error t value             Pr(>|t|)
## (Intercept) 0.471595   0.023821    19.8 < 0.0000000000000002 ***
## POWERDIFF   0.013976   0.001357    10.3   0.0000000000000104 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1845 on 58 degrees of freedom
## Multiple R-squared:  0.6464, Adjusted R-squared:  0.6403
## F-statistic:   106 on 1 and 58 DF,  p-value: 0.00000000000001035
```

Predicting with Ratings

Away team probability

```
modelA = lm(AW.1~POWERDIFF, data=freqs)
summary(modelA)
##
## Call:
## lm(formula = AW.1 ~ POWERDIFF, data = freqs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.73806 -0.06349 -0.02501  0.03762  0.40956
##
## Coefficients:
##             Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.329188   0.019875   16.56 <0.0000000000000002 ***
## POWERDIFF   -0.014099   0.001132  -12.45 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1539 on 58 degrees of freedom
## Multiple R-squared:  0.7277, Adjusted R-squared:  0.723
## F-statistic:   155 on 1 and 58 DF,  p-value: < 0.00000000000000022
```

Predicting with Ratings

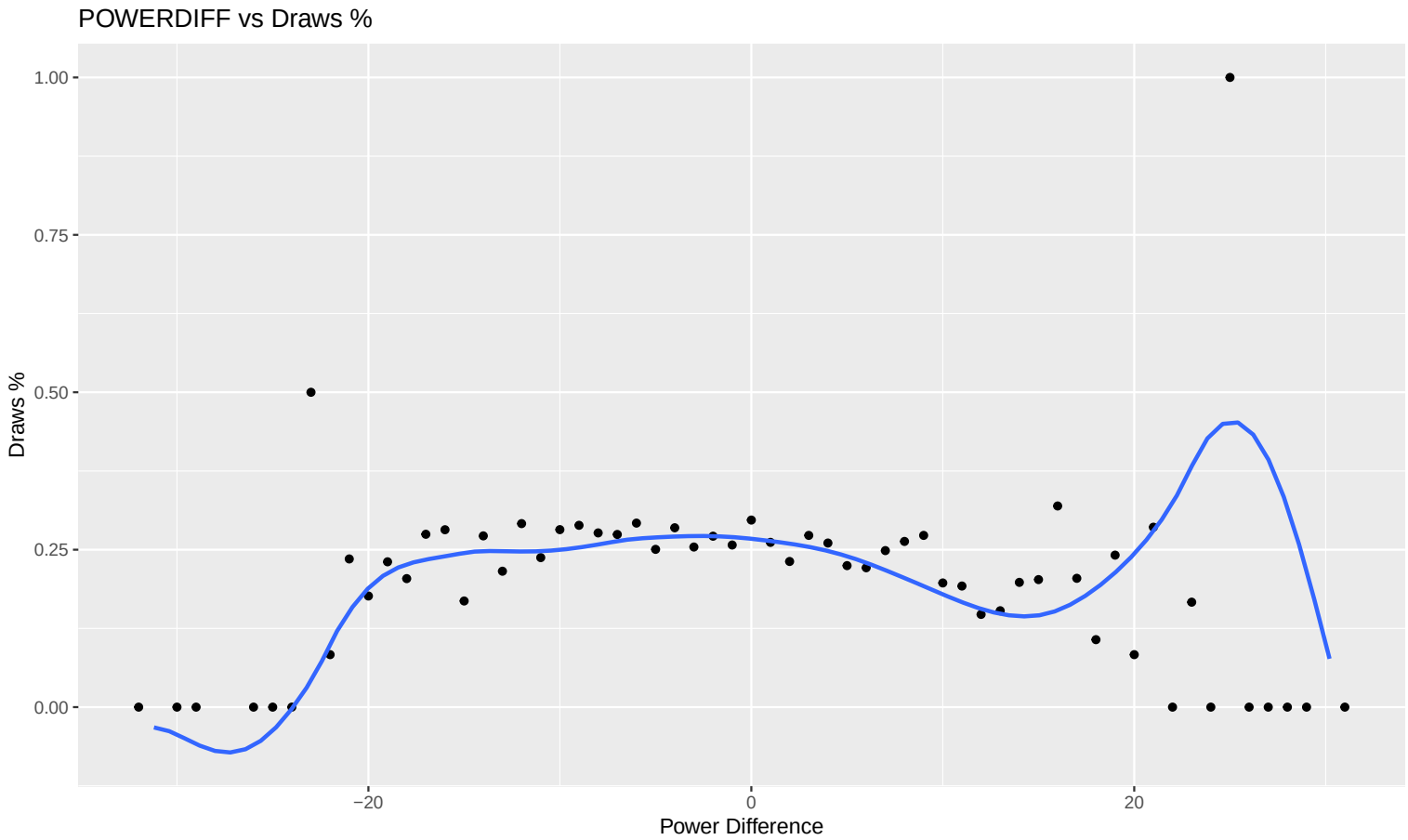
Draw: bad model

```
modelD = lm(D.1~POWERDIFF, data=freqs)
summary(modelD)
##
## Call:
## lm(formula = D.1 ~ POWERDIFF, data = freqs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.20302 -0.09901  0.02835  0.07346  0.79771
##
## Coefficients:
##             Estimate Std. Error t value           Pr(>|t|)
## (Intercept) 0.1992169  0.0203305   9.799 0.0000000000000655 ***
## POWERDIFF   0.0001228  0.0011584   0.106              0.916
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1574 on 58 degrees of freedom
## Multiple R-squared:  0.0001937,  Adjusted R-squared:  -0.01704
## F-statistic: 0.01124 on 1 and 58 DF,  p-value: 0.9159
```

Predicting with Ratings

Relationship with probability of draw is more like polynomial

```
ggplot(freqs, aes(x=POWERDIFF, y=D.1))+geom_point()+
  geom_smooth(formula=y~poly(x,2), se=F)+
  labs(x="Power Difference", y="Draws %", title = "POWERDIFF vs Draws %")
```



Predicting with Ratings

Draw: very hard to predict

```
modelD = lm(D.1~poly(POWERDIFF,2), data=freqs)
summary(modelD)
##
## Call:
## lm(formula = D.1 ~ poly(POWERDIFF, 2), data = freqs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.15926 -0.05704 -0.01525  0.01909  0.87538
##
## Coefficients:
##                     Estimate Std. Error t value             Pr(>|t|)
## (Intercept)          0.19927    0.01821  10.941 0.00000000000000126 ***
## poly(POWERDIFF, 2)1  0.01669    0.14108   0.118             0.906244
## poly(POWERDIFF, 2)2 -0.55058    0.14108  -3.903             0.000254 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1411 on 57 degrees of freedom
## Multiple R-squared:  0.211,  Adjusted R-squared:  0.1833
## F-statistic: 7.622 on 2 and 57 DF,  p-value: 0.001166
```

In-class assignment

In-class assignment

TIME FOR IN CLASS ACTIVITY !

- Pick an arbitrary football game from last season (`SEASON == 2026` in `f_data_sm`).
  - A mid-season game is better: both teams then have previous matches in the same season.
- **Distribution approach:** using only same-season games **before** the match you picked, estimate λ for the home and away teams. Build the Poisson score matrix (or use Skellam) and get P(home win), P(draw), P(away win).
- **Power difference approach:** compute last-6 goal supremacy for each team, then `POWERDIFF`. Read P(home win), P(draw), P(away win) from the frequency table or the fitted models.

In-class assignment

- Compare both sets of probabilities with
  - the **actual result** of the game
  - the **pre-game odds** (`H`, `D`, `A` in `f_data_sm` are Bet365 decimal odds; convert to probabilities if you like, $p = 1/\text{odds}$, optionally normalize so they sum to 1)
- Which approach was closer to the bookmaker? Which one matched the actual result?